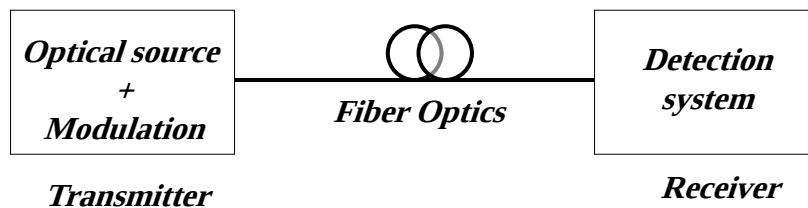

System Analysis

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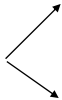
Point-to-Point Optical Communication System



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Design of a point to point optical Comm. system

- Design of optical link 
 - Performance
 - Cost

** Needed key system requirements to analyze a link :*

- 1- Desired (or possible transmission distance)
- 2- Data rate or channel bandwidth
- 3- Bit –error rate (BER)

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Design of a point to point optical Comm. system

- *To fulfill system requirements* → choose system components and their associated specifications.

First, choose operating wavelength :

- (800-to-900nm region) for relatively short distance
- (1300 or 1550 nm regions) for relatively long distance


lower dispersion lower attenuation

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Design of a point to point optical Comm. system

1- Optical Source

	<u>LED</u>	<u>Laser diode</u>
Spectral line width	Larger(~30nm)	Narrower(~0.1nm)
Output power	Lower	higher
Coupling efficiency with fiber	lower	higher
driver circuit	simpler	More complex
Cost	lower	higher
Number of emitting modes	higher	lower

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Design of a point to point optical Comm. system

2- photodiode

	<u>Pin</u>	<u>Avalanche</u>
Responsivity	lower	higher
Speed	-	-
Stability	higher	lower
Bias voltage	lower(~5V)	Higher(>40V)
Cost	lower	higher
Sensitivity	worse	better

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Design of a point to point optical Comm. system

3- optical fiber (Multimode or single mode)

- a) Core size
- b) Refractive index profile
- c) Bandwidth or dispersion
- d) Attenuation
- e) Numerical aperture or mode-field diameter

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Link Power Budget

- **Purpose** : To ensure that enough power will reach the receiver to maintain reliable performance during the entire system lifetime
- ***First*** , Calculate power margin (P_T) between optical transmitter output (P_S) and minimum receiver sensitivity (P_R) needed to establish a specified BER
- ***Second***, Allocate this margin to connector, splice, and fiber losses (attenuation) , plus (system margin) any additional margins required for possible component degradations, transmission-line impairments, or temperature effects.

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Link Power Budget

- Let: P_T = Total optical allowable loss in dB
 P_S = Optical transmitter output (source power) in dBm
 P_R = Minimum receiver sensitivity for specific BER in dBm
- $P_T = P_S - P_R$
 = Connector loss + fiber loss + coupling loss ++ System margin

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Link Power Budget

$$\text{Loss} = 10 \log P_{\text{out}} / P_{\text{in}} \quad \text{dB}$$

P_{in} , P_{out} are optical powers emanating into and out of loss element respectively

- For a very simple point to point system :

$$P_T = P_S - P_R$$

$$= 2L_C + \alpha_f L + \text{system margin}$$

L_C : connector loss (~0.5 → 1 dB), α_f : fiber attenuation (dB/km)

L: transmission distance, system margin (6 - 8 dB)

- Assumption: connectors only at the ends.
 : splice loss incorporated in cable loss.

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Link Power Budget

- For a very simple point to point system with multiple sections :

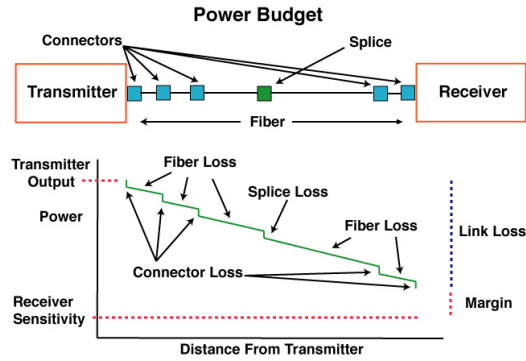
$$P_T = P_S - P_R = \sum \Gamma_C + \sum \alpha_{fi} L_i + \text{system margin}$$

Γ_C : Connector loss ($\sim 0.5 \rightarrow 1$ dB),

α_{fi} : fiber attenuation (dB/km) for section "i"

L_i : fiber length for the "i" section,

System margin $\sim 6 - 8$ dB



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Link Power Budget (design example)

- Requirements $\rightarrow$ Data rate : 20 Mb/s , BER : 10^{-9}
- Wavelength : 850 nm
- Photodetector : Si pin, sensitivity = -42 dBm
- Source : GaAlAs LED, emitted power = -13 dBm

$$P_T = P_S - P_R = -13 - (-42) = 29 \text{ dB}$$

Losses : 2 end connectors losses, 6 dB system margin,
fiber attenuation losses

$$P_T = 29 = 2(1 \text{ dB}) + \alpha_f L + 6 \text{ dB}$$

For $\alpha_f = 3.5$ dB/km, possible transmission path = 6 km

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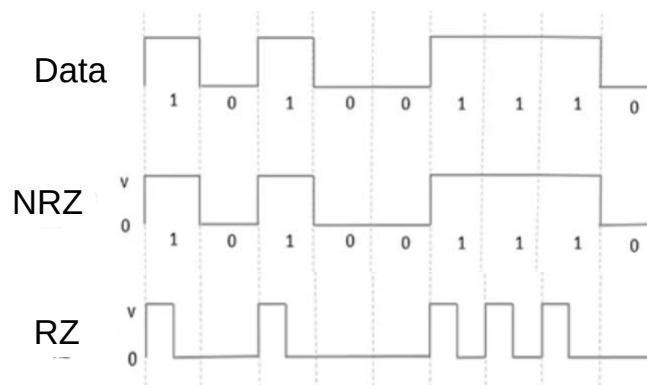
Non Return to Zero (NRZ) and Return to Zero (RZ) Formats

- Line coding is a technique which represents digital data into digital signals. Using this coding, it is possible to map sequence of bits into digital signal.
- The main digital coding formats used in optical communications are NRZ and RZ formats.
- For the NRZ format, the binary 1 is held for the whole bit period. **$\text{Bandwidth} = \text{Bitrate}/2$**
- For a RZ format the binary 1 is held for part (usually half) the bit period. **$\text{Bandwidth} = \text{Bitrate}$**

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Non Return to Zero (NRZ) and Return to Zero (RZ) Formats

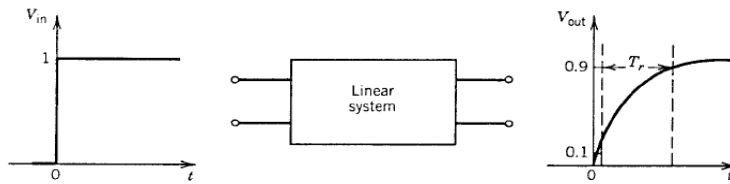


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Rise-Time Budget

- **Purpose** : To ensure that the system is able to operate properly at the intended bit rate
- **Rise time T_r of a linear system** : Time during which the response increases from 10 to 90% of its final output value when the input is changed abruptly.



Rise time T_r associated with a bandwidth-limited linear system.

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Example of the linear system –RC circuit

- $\Delta f \propto 1/T_r$ where Δf is the bandwidth
- When input voltage changes instantaneously from 0 to V_0 , output voltage changes as

$$V_{out}(t) = V_0[1 - \exp(-t/RC)] \quad (1)$$

$$, T_r = (\ln 9)RC \approx 2.2RC.$$

Taking Fourier Transform of (1)

$$H(f) = (1 + i2\pi fRC)^{-1}$$

Bandwidth Δf corresponds to the frequency at which $|H(f)|^2 = 1/2$

$$\Rightarrow \Delta f = (2\pi RC)^{-1}$$

$$T_r = 2.2/2\pi\Delta f = 0.35/\Delta f \quad (2)$$

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Rise-Time Budget

- $T_r \Delta f = 0.35$ could be used in design of optical communication systems
- Relationship between bandwidth Δf and bit rate B depends on digital format :

1- For return-to-zero (RZ) format

$$\Delta f = B, \quad B T_r = 0.35$$

2- For nonreturn-to-zero (NRZ) format

$$\Delta f \approx B/2, \quad B T_r = 0.7$$

$$\therefore T_r \leq \begin{cases} 0.35/B & \text{for RZ format,} \\ 0.70/B & \text{for NRZ format.} \end{cases}$$

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Rise-Time Budget

- $T_r^2 = T_{tr}^2 + T_{fiber}^2 + T_{rec}^2$
where T_{tr} , T_{fiber} , and T_{rec} are rise times associated with transmitter, fiber, and receiver, respectively
- $T_{tr} \Rightarrow$ determined primarily by optical source bandwidth, electrical parasitics associated with optical source, and electronic components of driving circuit

$$T_{tr} \begin{cases} \text{few nanoseconds for LED-based transmitters} \\ \text{can be shorter than 0.1 ns for laser-based transmitters} \end{cases}$$

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Rise-Time Budget

- $T_{\text{rec}} \Rightarrow$ determined primarily by 3-dB electrical bandwidth of receiver front end.

Equation (2) can be used to estimate T_{rec} if the front-end bandwidth is specified

- $T_{\text{fiber}}^2 \Rightarrow T_{\text{fiber}}^2 = T_{\text{modal}}^2 + T_{\text{chromatic}}^2$

For single-mode fibers $\Rightarrow T_{\text{modal}} = 0$ and $T_{\text{fiber}} = T_{\text{chromatic}}$

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Rise-Time Budget

- T_{modal} can be approximated by the time delay ΔT

$$T_{\text{modal}} \approx (n_1 \Delta / c) L,$$

where $\Delta = (n_1 - n_2) / n_1$

- $T_{\text{chromatic}} \approx |D| L \Delta \lambda$
where $\Delta \lambda$ is the spectral width of the optical source

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Rise-Time Budget (example)

- 1.3- μm lightwave system designed to operate at 1 Gb/s over a single-mode fiber with a repeater spacing of 50 km
- $T_{\text{tr}} = 0.25 \text{ ns}$, $T_{\text{rec}} = 0.35 \text{ ns}$, source spectral width $\Delta\lambda = 3 \text{ nm}$, average value of $D = 2 \text{ ps}/(\text{km}\cdot\text{nm})$ at the operating wavelength

$$\therefore T_{\text{Chrom}} = 0.3 \text{ ns} \quad \text{for a link length } L = 50 \text{ km}$$
$$, \quad T_{\text{moda}} = 0 \quad , \quad T_{\text{fiber}} = 0.3 \text{ ns}$$

$$\therefore T_r = 0.524 \text{ ns}$$

$\Rightarrow$ It cannot be operated at 1 Gb/s with RZ format
but can be operated at 1 Gb/s with NRZ format

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